

# Overtune 3

## Contributors Guide & DSP Knowledge Base

Architecture Specifications, Development Standards, and Canonical DSP Literature

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### Abstract

This guide provides developers and DSP researchers with the architectural specifications, contribution workflows, coding guidelines, and academic references needed to contribute to **Overtune 3**. The codebase combines a zero-dependency, pure C++20 digital signal processing core with a reactive, hardware-accelerated Qt 6 / QML user interface.

## 1 System Architecture & Repository Layout

Overtune 3 is partitioned into two decoupled modules to maximize performance and portability:

1. **dsp/ Module:** Pure, self-contained C++20 digital signal processing library. Contains zero dependencies on GUI, Qt, or external linear algebra libraries. Uses standard C++ `<complex>`, `<vector>`, and `<cmath>`.
2. **app/ Module:** High-performance Qt 6.4+ / QML desktop application featuring custom hardware-accelerated Canvas plotters, interactive pole-zero manipulation, real-time audio playback, and responsive layout management.

### 1.1 Directory Structure

```
overtune3/
|-- dsp/
|   |-- include/dsp/          # Public DSP headers (Biquad, FilterTypes, Cascade)
|   |-- src/                  # Pure C++20 synthesis implementations
|       |-- Butterworth.cpp
|       |-- Chebyshev.cpp
|       |-- Elliptic.cpp
|       |-- Bessel.cpp
|       |-- BiquadCascade.cpp
|-- app/
|   |-- src/                  # Qt/QML C++ controllers (FilterEngine, Models)
|   |-- qml/                  # Modern declarative UI components & pages
|       |-- components/      # Reusable controls (Plots, Buttons, SegmentedBar)
|       |-- pages/           # Workspaces (Design, Analysis, Simulation, Export)
|       |-- theme/           # Design tokens, fonts, and dark/light color palettes
|   |-- resources/           # Vector icons, bundled fonts, and equation assets
|-- latex/                   # Comprehensive LaTeX documentation sources & PDFs
```

## 2 Contribution Standards and Guidelines

### 2.1 Core DSP Implementation Principles

When contributing new filter approximations, equalizer models, or analysis algorithms to `dsp/`:

- **Zero Heap Allocations in Audio Loops:** The inner processing loop must operate purely on pre-allocated buffers or stack arrays with deterministic execution times.
- **Numerical Stability:** Always decompose polynomials of order  $N > 2$  into Second-Order Sections (SOS). Pair complex conjugate poles with their nearest transmission zeros to maximize dynamic range and minimize quantization noise.
- **Frequency Pre-Warping:** Always pre-warp continuous-time cutoff frequencies using  $\Omega = 2F_s \tan(\pi F_c / F_s)$  before applying the bilinear transform.
- **SIMD-Friendly Loops:** Structure Direct Form II Transposed loops to enable compiler auto-vectorization across contiguous audio sample blocks.

### 2.2 Building and Testing the Application

To compile and test the entire project locally on Linux/macOS:

```
git clone https://github.com/shadcy/overtune3.git
cd overtune3
./build.sh # Release build in build/
./build/bin/FilterDesigner # Launch application
```

## 3 Canonical DSP Reference Literature

The mathematical algorithms implemented in Overtune 3 are rooted in foundational digital signal processing literature:

1. **Alan V. Oppenheim & Ronald W. Schaffer:** *Discrete-Time Signal Processing*, 3rd Edition, Prentice Hall. (Foundational treatment of Bilinear Transform, Z-transforms, and IIR stability).
2. **John G. Proakis & Dimitris G. Manolakis:** *Digital Signal Processing: Principles, Algorithms, and Applications*, Pearson. (Comprehensive derivations of Butterworth, Chebyshev, and Elliptic analog prototypes).
3. **Udo Zölzer:** *DAFX: Digital Audio Effects*, 2nd Edition, John Wiley & Sons. (Practical audio filter implementations, biquad structures, and perceptual equalizers).
4. **Robert Bristow-Johnson:** *Cookbook formulae for audio equalizer biquad filter coefficients*. (Standard reference for second-order parametric audio shelving and peaking filters).

## 4 Community & External Links

- [Overtune 3 Project Repository on GitHub](#)
- [Overtune 3 Issue Tracker & Feature Requests](#)
- [Project Maintainer Profile — shadcy](#)
- [SciPy Signal Processing Library Documentation](#)
- [Digital Signal Processing Overview — Wikipedia](#)
- [Digital Biquad Filter Topologies — Wikipedia](#)